

Thm. If $\lim_{x \rightarrow a} g(x) = m$, and $m \neq 0$.
then $\lim_{x \rightarrow a} \frac{1}{g(x)} = \frac{1}{m}$

Roughwork

Given $\epsilon > 0$, $\begin{cases} \text{for } x \text{ close enough to } a \\ |g(x) - m| \text{ very small} \end{cases}$

$$\left| \frac{1}{g(x)} - \frac{1}{m} \right| = \left| \frac{m - g(x)}{g(x)m} \right|$$

We know that very close to a , $g(x)$ is about $m \neq 0$

So $g(x)$ is about $m^2 \neq 0$

Pf. Let $\epsilon > 0$.

Since $\lim_{x \rightarrow a} g(x) = m$

For any $\epsilon_0 > 0$, $\exists \delta$ s.t.

$0 < |x - a| < \delta$, we have $|g(x) - m| < \epsilon_0$

Also we have

$$\left| \frac{1}{g(x)} - \frac{1}{m} \right| = \left| \frac{m - g(x)}{g(x)m} \right| < \frac{\epsilon_0}{|g(x)m|}$$

for $0 < |x - a| < \delta$

*. How do we approximate $\frac{1}{|g(x)m|}$?

By $\triangle$ ineq,

$$|m| - |g(x)| \leq |m - g(x)|$$

Rearrange,

$$\frac{1}{|m| - |m - g(x)|} \geq \frac{1}{|g(x)|}$$

If x is very close to a , $|g(x)|$ is close to $|m| \neq 0 \Rightarrow |g(x)| \neq 0$

$|m - g(x)|$ arbitrarily small, $|m| \neq 0 \Rightarrow |m| - |m - g(x)| \neq 0$

$$|m| - |g(x)| \leq |m - g(x)| < \varepsilon_0$$

$$|m| - \varepsilon_0 \leq |g(x)| \Rightarrow \frac{1}{|m| - \varepsilon_0} \geq \frac{1}{|g(x)|}$$

Then (*) becomes

$$\left| \frac{1}{g(x)} - \frac{1}{m} \right| < \frac{\varepsilon_0}{(|m| - \varepsilon_0)|m|}$$

Want $< \varepsilon$

Solve for ε_0 :

$$\frac{\varepsilon_0}{|m|(|m| - \varepsilon_0)} = \varepsilon$$

$$\varepsilon_0 = \varepsilon |m| (|m| - \varepsilon_0)$$

$$\varepsilon_0 (1 + \varepsilon |m|) = \varepsilon |m|^2$$

$$\varepsilon_0 = \frac{\varepsilon |m|^2}{1 + \varepsilon |m|}$$



Q. How can we define

$$\lim_{x \rightarrow a^-} f(x) = l ?$$

$$\lim_{x \rightarrow a^+} f(x) = l.$$

$$\forall \varepsilon > 0, \exists \delta > 0, \text{ s.t.}$$

$$0 < a - x < \delta \text{ then}$$

$$|f(x) - l| < \varepsilon.$$

Note If $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a} f(x) = l$

Q. How can we define $\lim_{x \rightarrow \infty} f(x) = l$?

$$\forall \varepsilon > 0, \exists M > 0 \text{ s.t.}$$

$$\text{Whenever } x > M, |f(x) - l| < \varepsilon.$$

Show $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$

PF. Let $\varepsilon > 0$, Want a $M > 0$, s.t. whenever $x > M$,

$$\left| \frac{1}{x} - 0 \right| < \varepsilon$$

$$\left| \frac{1}{x} \right| < \frac{1}{M}$$

$$\text{Solve for } \frac{1}{M} = \varepsilon \Rightarrow M = \frac{1}{\varepsilon}.$$



Def. A function f is continuous at a if $\lim_{x \rightarrow a} f(x) = f(a)$.

$\forall \varepsilon > 0, \exists \delta > 0$, s.t.

whenever $0 < |x - a| < \delta$,

then $|f(x) - f(a)| < \varepsilon$.

Thm. Let f, g be functions continuous at a .

then

① $f+g$ is also cts. at a .

② $f \cdot g$ is cts. at a .

③ if $g(a) \neq 0$, then $\frac{1}{g(x)}$ cts. at a .

PF. ① Have proven $\lim_{x \rightarrow a} (f+g)(x) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$
 $= f(a) + g(a)$ (b/c f, g given to be cts. at $x=a$).
 $= (f+g)(a)$.

② ③ Exercise.



Ex. let $f(x) = \begin{cases} x \sin \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

As at $x=0$?

Df. Let $\varepsilon > 0$, want $\delta > 0$. s.t.

whenever $0 < |x| < \delta$

want $|x \sin \frac{1}{x}| < \varepsilon$

$$|x \sin \frac{1}{x}| \leq |x| < \varepsilon$$

let $\delta = \varepsilon$, then done.

